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Differential Amplifier Simulation

This notebook simulates the operation of a differential amplifier. It includes illustrations, diagrams, and explanations for each code block.

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import lti, step
```

Define Differential Amplifier Parameters

Parameters:

- **A_d**: Differential mode gain (Amplifies the difference between inputs)
- **A_c**: Common mode gain (Ideally should be minimized for noise rejection)
- **CMRR**: Common-mode rejection ratio, defined as the ratio of A_d to A_c

```
In [2]: A_d = 10 # Differential mode gain
A_c = 0.1 # Common mode gain
CMRR = A_d / A_c # Common-mode rejection ratio
```

Generate Input Signals

- **V1**: Sine wave input representing one signal source
- **V2**: Sine wave with half the amplitude, representing the second input
- **T**: Time vector for plotting the signals

```
In [3]: T = np.linspace(0, 1, 1000) # Time vector
V1 = np.sin(2 * np.pi * 10 * T) # Input signal 1
V2 = 0.5 * np.sin(2 * np.pi * 10 * T) # Input signal 2 (different amplitude)
```

Compute Differential and Common-Mode Signals

- **V_{diff}**: Difference between V1 and V2 (true signal to be amplified)
- **V_{common}**: Average of V1 and V2 (undesired noise component)

```
In [4]: V_diff = V1 - V2
V_common = (V1 + V2) / 2
```

Compute Amplifier Output

- **V_out_diff**: Amplified differential mode output
- **V_out_common**: Common-mode output contribution
- **V_out_total**: Final amplifier output including both modes

```
In [5]: V_out_diff = A_d * V_diff
        V_out_common = A_c * V_common
        V_out_total = V_out_diff + V_out_common
```

Plot Results

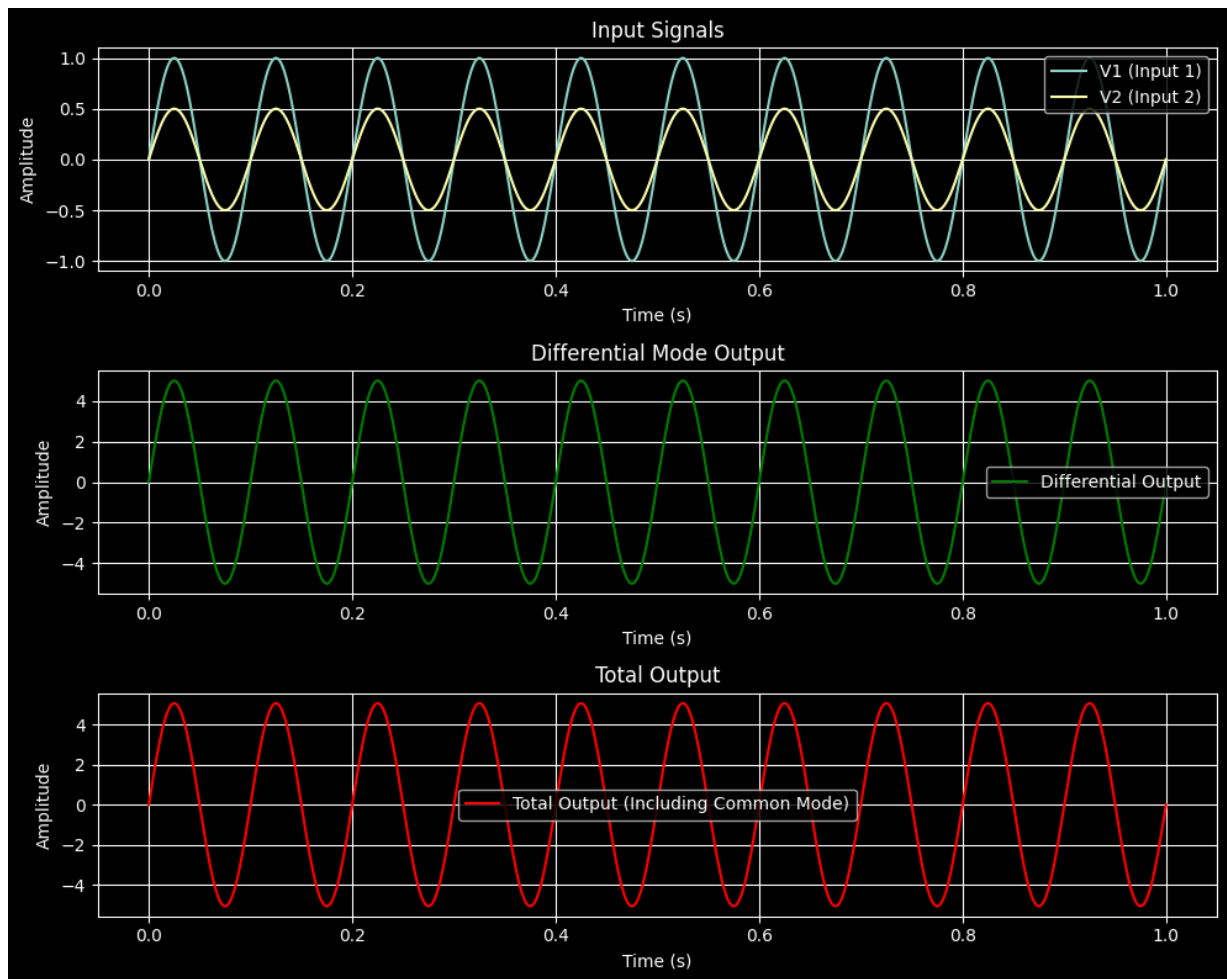
```
In [6]: plt.figure(figsize=(10, 8))

# Plot input signals
plt.subplot(3, 1, 1)
plt.plot(T, V1, label='V1 (Input 1)')
plt.plot(T, V2, label='V2 (Input 2)')
plt.title('Input Signals')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.legend()
plt.grid()

# Plot differential mode output
plt.subplot(3, 1, 2)
plt.plot(T, V_out_diff, label='Differential Output', color='g')
plt.title('Differential Mode Output')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.legend()
plt.grid()

# Plot total output
plt.subplot(3, 1, 3)
plt.plot(T, V_out_total, label='Total Output (Including Common Mode)', color='r')
plt.title('Total Output')
plt.xlabel('Time (s)')
plt.ylabel('Amplitude')
plt.legend()
plt.grid()

plt.tight_layout()
plt.show()
```



Display Common-Mode Rejection Ratio

- A high **CMRR** indicates better noise rejection in practical circuits.

```
In [7]: print(f'Common-Mode Rejection Ratio (CMRR): {CMRR:.2f}')
```

Common-Mode Rejection Ratio (CMRR): 100.00